Sorts of Distributed Systems

Distributed Systems

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Next in Line...

Which Sorts of Distributed Systems Around?

- distributed systems are *not* a new thing
- decades ago, distributed computing was a thing even before we started understanding what computation actually is
- nowadays, distributed systems are anywhere: their sorts are so diverse in nature, structure, organisation, dynamics, that classifying them all is possibly neither an easy task nor a useful one
- however, in the first decades of their existence, the mutable needs and available resources promoted an *evolution* of distributed systems
- which basically generated over time three recognisable *fundamental classes* of distributed systems

Disclaimer

- the following slides borrow a great deal from the slides coming with the book adopted as the basic one for this course [Tanenbaum and van Steen, 2017]
- material from those slides (including pictures) has been re-used in the following, and integrated with new material according to the personal view of the professor
- every problem or mistake contained in these slides, however, should be attributed to the sole responsibility of this course's professor

Evolution of Distributed Systems

Evolving technological and application scenarios. . .

- . . . forced new requirements, structure, and goals for distributed systems
- resulting in the emergence of diverse classes of distributed systems over time

Sorts of Distributed Systems

Three main classes of distributed systems

- distributed computing systems
- distributed information systems
- distributed pervasive systems

Next in Line...

Distributed Computing Systems

The main characteristic

- using a *multiplicity of distributed computers* to perform *high-performance* tasks

Two classes

- cluster computing systems
- grid computing systems

Cluster Computing Systems I

The basic idea

- a collection of similar workstations / PCs
- running the same OS
- located in the same area
- interconnected through a high-speed LAN

Motivation

- the ever increasing price / performance ration of computers makes it cheaper to build a supercomputer by putting together many simple computers, rather than buying a high-performance one
- also, robustness is higher, maintenance and incremental addition of computing power is easier

Cluster Computing Systems II

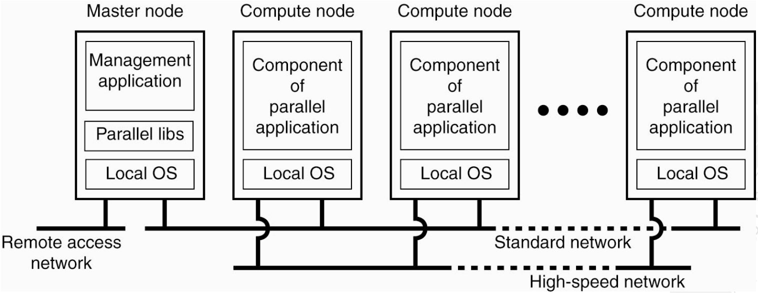
Usage

- parallel computing
- typically, a single computationally-intensive program is run in parallel on multiple machines

Cluster Computing Systems: Example

Beowulf clusters

- Linux-based
- each cluster is a collection of computing nodes controlled and accessed by a *single* master *node*



[Tanenbaum and van Steen, 2017]

Cluster vs. Grid Computing Systems

Homogeneity vs. heterogeneity

- homogeneity
  - computers in a cluster are typically similar
  - computers in a cluster have the same OS
  - computers in a cluster are connected to the same (local) network
- in essence, cluster computer systems are homogeneous
- grid computer systems instead are typically heterogeneous

Grid Computing Systems

The main idea

- resources from different organisations are brought together to promote *collaboration* between individuals, groups, or institutions, by passing organisation boundaries
- collaboration is built in the form of a *virtual organisation*
  - essentially, a new virtual organisational entity including people from existing organisations
  - accessing resources made available by participating organisations
  - including servers, databases, hard disks, ...
- by their very nature, grid computer systems deal with different administrative domains

Architecture of a Grid Computing System I

A layered architecture for a grid computing system [Foster et al., 2001]

fabric layer — interface to local resources at a specific site

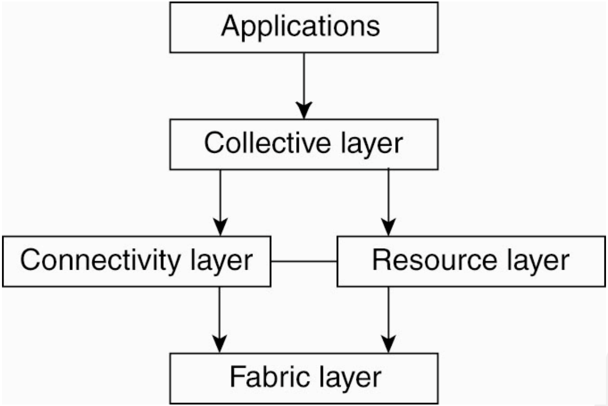
resource layer — management of single resources — e.g., access control

connectivity layer — communication protocols for grid transactions spanning over multiple resources, plus security protocols for authentication

collective layer — handling access to multiple resources—resource discovery, allocation, ...

application layer — applications operating in the virtual organisation

Architecture of a Grid Computing System II



[Tanenbaum and van Steen, 2017]

Architecture of a Grid Computing System III

Grid middleware layer

- the *core* of a grid middleware layer is represented by *connectivity*, *resource*, and *collective* layers
- altogether, they provide *uniform access* to otherwise dispersed *resources*

Next in Line...

Distributed Information Systems

Origin

- many separate networked applications to be integrated
- structural problems of interoperability

Sorts

- several non-interoperating servers shared by a number of clients: distributed queries, distributed transactions

→ Transaction Processing Systems

- several sophisticated applications – not only databases, but also processing components – requiring to directly communicate with each other

→ Enterprise Application Integration (EAI)

Transaction Processing Systems

Distributed transactions for distributed databases

- operations on databases are usually performed in terms of transactions
- when databases are distributed, transactions should be *distributed*
- *special primitives* from the distributed system or the runtime system

ACID properties

atomic steps of a transaction occurs *invisibly* to the outside world
consistent the transaction does *not violate* system *invariants*
isolated *concurrent* transactions do *not interfere* with each other
durable once a transaction *commits*, its effects are *permanent*

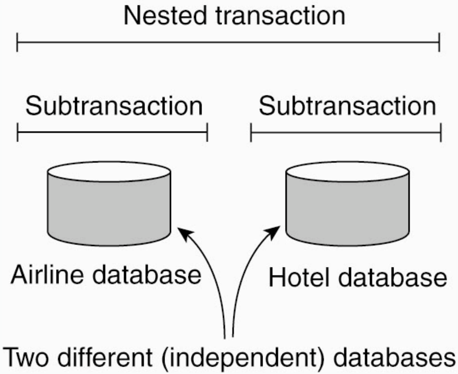
Example Primitives for Transactions

Primitive	Description
BEGIN_TRANSACTION	Mark the start of a transaction
END_TRANSACTION	Terminate the transaction and try to commit
ABORT_TRANSACTION	Kill the transaction and restore the old values
READ	Read data from a file, a table, or otherwise
WRITE	Write data to a file, a table, or otherwise

[Tanenbaum and van Steen, 2017]

Nested Transactions

A nested transaction is made of a number of subtransactions



[Tanenbaum and van Steen, 2017]

Nesting in transactions could be arbitrarily deep

The Problem with Nested Transactions I

Durability of nested and sub-transactions

- a *whole* nested transaction should exhibit ACID properties
- so, if a subtransaction fails, all subtransactions till there should be *undone*, even though they already committed
- the effects of subtransactions could not be really durable if the whole transanction does not succeed

→ *durability* here refers to the *top-level* transaction

The Problem with Nested Transactions II

Private copy of the world as a solution

- all transactions are performed over a copy of the data, so subtransactions could keep ACIDity in the *local world*
- the effect of a *successful* nested transaction would be *propagated* only *after* it *succeeds*

→ in case, the *copy* of the world transformed *becomes* the *world*

→ in any case, transactions are ACID

Nested Transactions in Distributed Information Systems

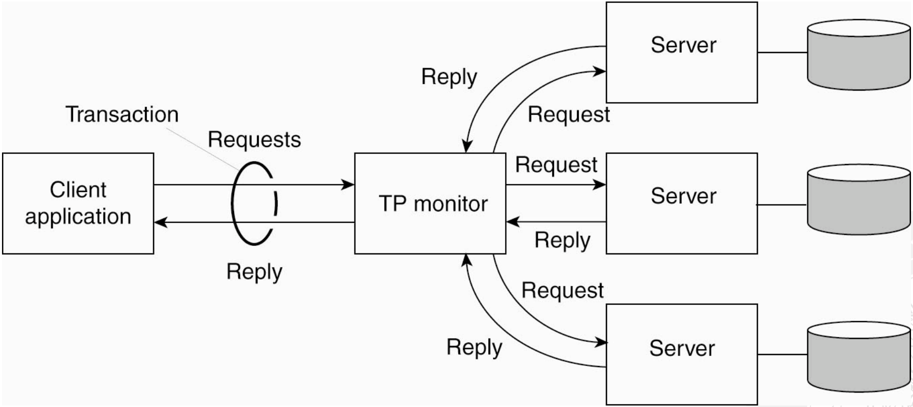
Nested transactions are a natural way for distributing transactions

- "leaf" subtransactions are usual transactions over single servers
- distributed transactions are nested transactions

An early solution: TP monitor

- transaction processing monitor (or, *TP monitor*)
- to allow applications to access *multiple* DB *servers*
- with a *transactional semantics*

TP Monitor



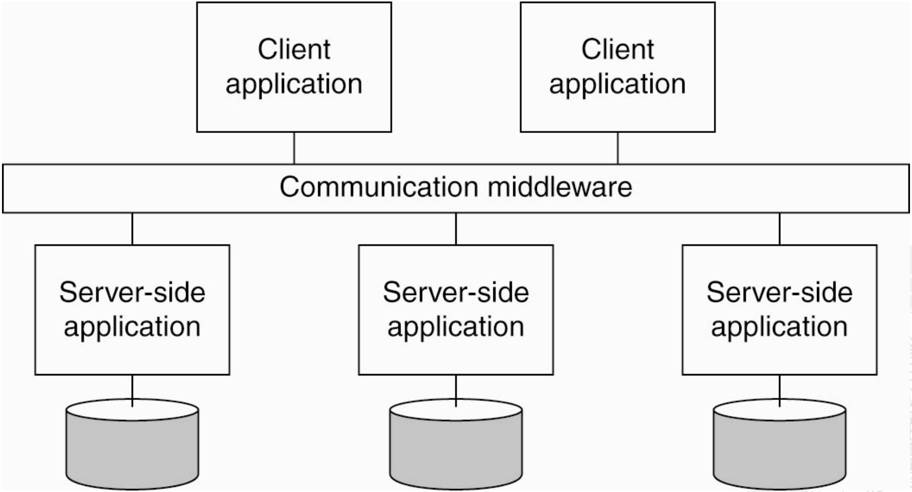
[Tanenbaum and van Steen, 2017]

Enterprise Application Integration

It is not only a matter of accessing distributed databases

- integration should happen at the application level, too
- beyond data integration, process integration
- application should interact and communicate meaningfully with each other

Middleware as a Communication Facilitator



[Tanenbaum and van Steen, 2017]

Next in Line. . .

Distributed Systems with Instability

What happens when instability is the default condition?

- ... like, with mobile devices with batteries and sporadic network connection?
- ... like, in modern *distributed pervasive systems*?

Main features

- a distributed pervasive system is *part of* our *surroundings*
- a distributed pervasive system generally *lacks* of a *human* administrative *control*

Requirements for Pervasive Systems

Three requirements[Grimm et al., 2004]

- embrace contextual changes
- encourage ad hoc composition
- recognise sharing as the default

Remarks

- a device must be continually *aware* of the fact that its *environment may* change at any time
- many devices in pervasive system will be *used in different ways* by different users
- devices generally join the system in order to access (provide) information: *information* should then be easy to read, store, manage, and *share*

Home Systems

Systems built around home networks

- no way to ask people to act as a competent network / system administrator

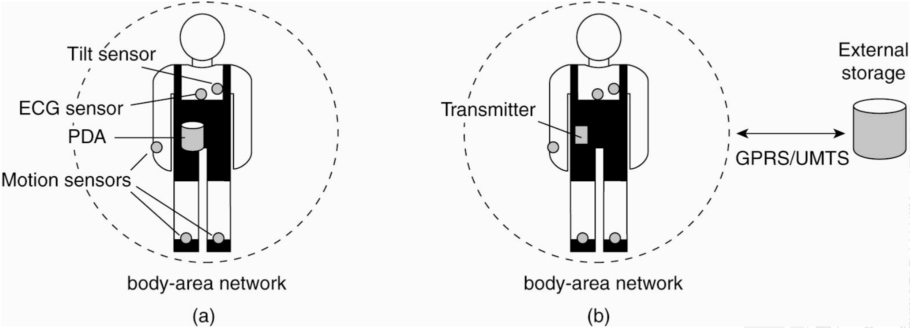
→ home systems should be self-configuring and self-maintaining in essence

Systems built around personal information

- *huge amount* of heterogeneous personal *information* to be managed,
- coming from *heterogeneous sources* from inside and outside the home system

Health Care Systems

- Personal systems built around a Body Area Network
- Possibly, minimising impact on the person—like, preventing free motion



[Tanenbaum and van Steen, 2017]

Health Care Systems: Questions to be Addressed

- where and how should monitored *data* be *stored*?
- how can we prevent *loss* of crucial *data*?
- what infrastructure is needed to generate and propagate *alerts*?
- how can physicians provide online *feedback*?
- how can extreme *robustness* of the monitoring system be realised?
- what are the *security* issues and how can the proper policies be enforced?

Sensor Networks

An enabling technology for pervasive systems

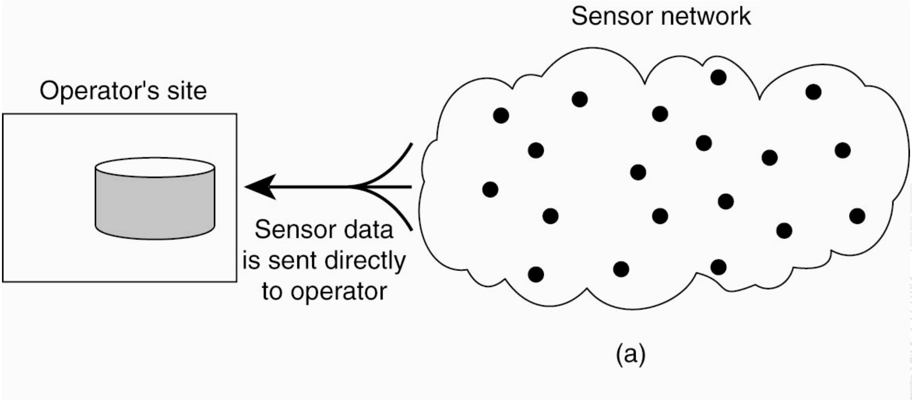
- clouds of *spatially-distributed sensors*—from tens to thousands of nodes with a sensing device
- acquiring, processing and transmitting *environmental information*

A possible view: distributed databases

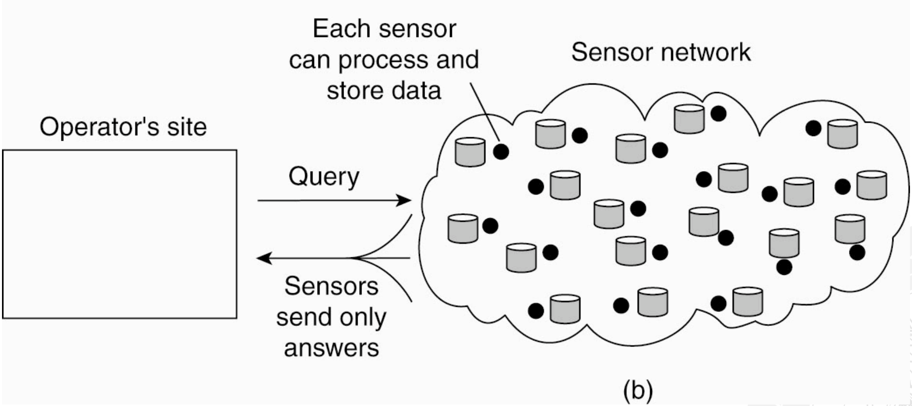
- distributed sources of information
- that can possibly be queried along time
- two possible extremes: either sensors just send information in without cooperating, or they do all the computation and return the results

↑ both solutions are bad ones, since they require either too much network consumption or too much node power consumption

Architecture of Sensor Networks: Two Extremes I



Architecture of Sensor Networks: Two Extremes II



[Tanenbaum and van Steen, 2017]

Sensor Networks: A Solution

In-network data-processing

- setting up a tree network among sensors
- passing queries through the sensor tree
- aggregating the results at the different levels of the tree

Questions to be addressed

- how do we (dynamically) set up an efficient tree in a sensor network?
- how does aggregation of results take place? Can it be controlled?
- what happens when network links fail?

Next in Line. . .

Components & Infrastructure I

Complexity & Components

- the increasing complexity of today distributed systems is only manageable by groups of engineers and practitioners
- *community efforts* are essential, in particular for critical *infrastructure components*
  - open software as the default
- collections of *components* to provide foundation for the design of complex distributed systems are available
- it is fundamental to
  - keep an eye on the community projects available
  - participate to the community process so as to know the state of each project, and possibly contribute
  - understand the best selection of components for our systems

Components & Infrastructure II

Some sources: Foundations

- The Apache Software Foundation (ASF) https://www.apache.org
- The Cloud Native Computing Foundation (CNCF) https://www.cncf.io/
- The Free Software Foundation (FSF) https://www.fsf.org/
- The Python Software Foundation (PSF) https://www.python.org/psf-landing/
- ...

The Cloud Native Computing Foundation (CNCF) I

About CNCF

- part of the Linux Foundation https://linuxfoundation.org/
- collects and promotes projects as the foundation of *cloud native computing*
  - graduated / incubating / sandbox / archived projects
  - depending on the different stage of development
- including Kubernetes, etc, Prometheus

The Cloud Native Computing Foundation (CNCF) II

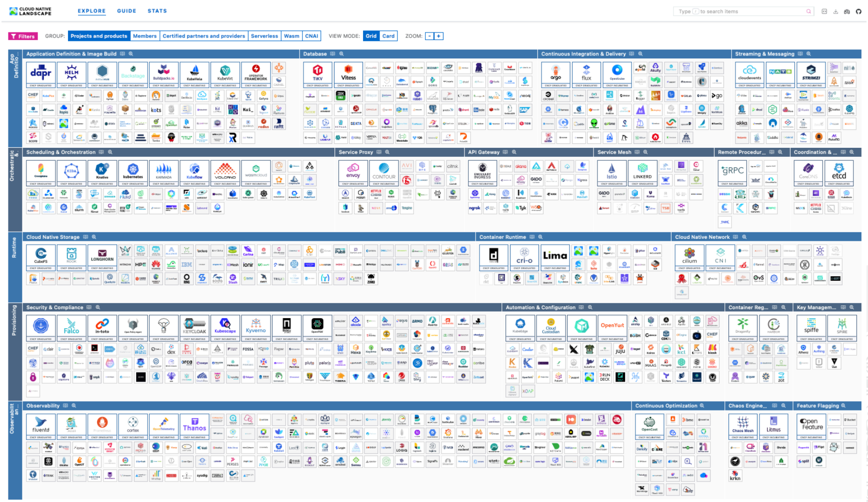
From the *CNCF Charter*

Cloud native technologies empower organizations to build and run scalable applications in modern, dynamic environments such as public, private, and hybrid clouds. Containers, service meshes, microservices, immutable infrastructure, and declarative APIs exemplify this approach.

These techniques enable loosely coupled systems that are resilient, manageable, and observable. Combined with robust automation, they allow engineers to make high-impact changes frequently and predictably with minimal toil.

The Cloud Native Computing Foundation seeks to drive adoption of this paradigm by fostering and sustaining an ecosystem of open source, vendor-neutral projects. We democratize state-of-the-art patterns to make these innovations accessible for everyone.

The Cloud Native Computing Foundation (CNCF) III



https://landscape.cncf.io

Next in Line...

Summing Up I

Diverse sorts of distributed systems exist

- roughly telling the story of the *evolution* of distributed systems over the years
    - distributed computing systems
    - distributed information systems
    - (distributed) pervasive systems
- depending on both the *environment* where they are developed, the *goals* they have to achieve, and the level of the available *technologies*
- different *models*, *methodologies*, and *technologies* are to be used to design and develop different sorts of distributed systems
- today, complexity of distributed systems makes them typically borrow features from *all* the three basic classes

Summing Up II

Nowadays. . .

- the complexity of distributed systems engineering calls for community efforts
- hundreds of (critical) software components are available to be looked up, selected, and used

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